Lecture 4: Data Representation III — Struct Layout & Alignment

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Warm-Up: Printing a Number in Binary

C has no `%b` conversion. If you want to see the bits, you write it yourself — and everything you need is from Lecture 2:

```c
void print_binary(uint32_t x)
{
    for (int i = 31; i >= 0; i--) {
        putchar((x & (1u << i)) ? '1' : '0');
        if (i % 4 == 0 && i != 0)
            putchar(' ');        // one space per nibble = one hex digit
    }
}
```

- `1u << i` builds a **mask** — one bit set, at position `i`. `1u`, not `1`: `1 << 31` is undefined on a 32-bit `int`.
- `x & mask` **tests** that bit: nonzero if set, zero if not.

- Counting down from 31 prints the high bit first, the way we write numbers.

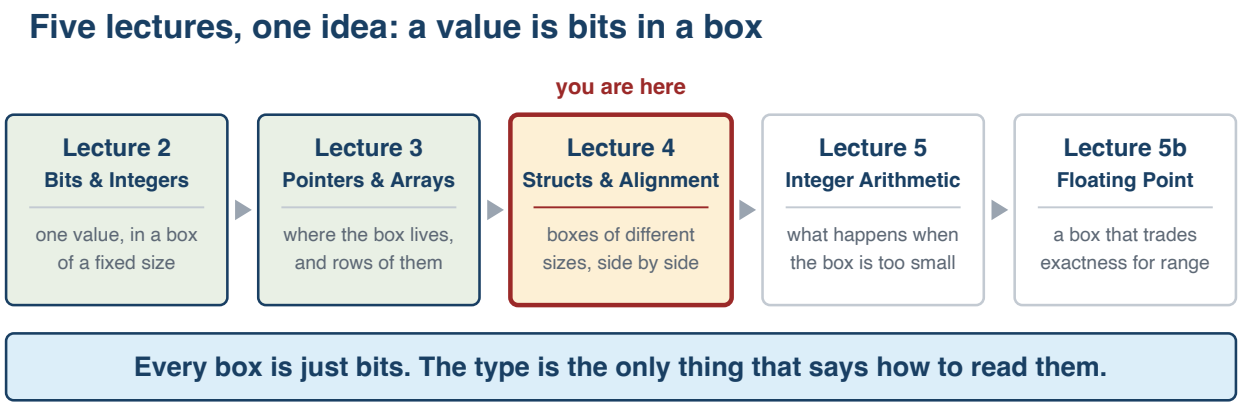
Run It: Old Friends in Binary

```
5               00000005  0000 0000 0000 0000 0000 0000 0000 0101
42 (0x2A)       0000002A  0000 0000 0000 0000 0000 0000 0010 1010
1u << 31        80000000  1000 0000 0000 0000 0000 0000 0000 0000
0xFF00FF00      FF00FF00  1111 1111 0000 0000 1111 1111 0000 0000
-1              FFFFFFFF  1111 1111 1111 1111 1111 1111 1111 1111
-5              FFFFFFFB  1111 1111 1111 1111 1111 1111 1111 1011
INT32_MIN       80000000  1000 0000 0000 0000 0000 0000 0000 0000
(char)-5 -> int FFFFFFFB  1111 1111 1111 1111 1111 1111 1111 1011
```

Read `FF00FF00` against the groups of four — each hex digit **is** one group. `1u << 31` and `INT32_MIN` are the same pattern, and so are the last two lines: that is sign extension, widening a `char` without changing what it means.

Keep this function. You will want it from here on.

Where We Are



Lecture 2 filled in one box; Lecture 3 gave it an address. Today: boxes of **different sizes**, side by side — and the gaps the machine leaves between them.

Three Arrays, or One?

A student has a name, an ID and a GPA — so far, three arrays you keep in step by hand. Sort one, forget the others, and the roster is silently wrong:

```c
char names[100][16];   int ids[100];   float gpas[100];
```

```c
struct Student {
    char  name[16];
    int   id;
    float gpa;
};

struct Student roster[100];      // one array now, not three
roster[7].id = 1001;             // reach a field with a dot
```

`struct Student` is a **type**, like `int`.

`typedef`: Dropping the Word `struct`

Writing `struct Student` everywhere gets old. `typedef` gives the type a second name:

```
typedef struct {
    char  name[16];
    int   id;
    float gpa;
} Student;

Student roster[100];        // no "struct" needed
```

Most C code you read does this. It changes nothing about the layout — the same padding rules apply, byte for byte.

Filling One In

Three ways, and the third is the one to prefer:

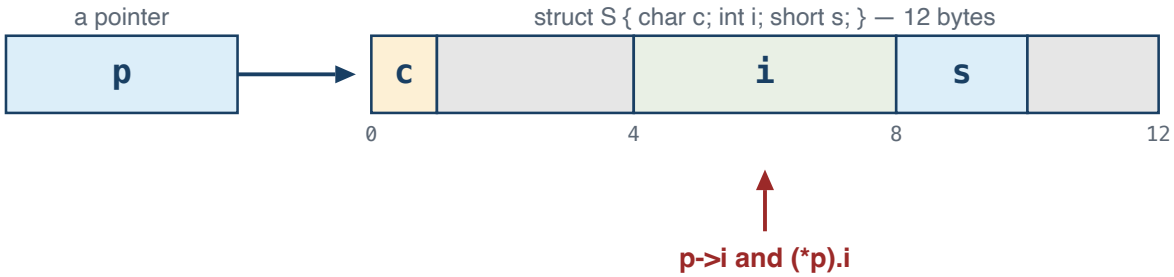
```
struct Student b = { "Beto", 1002, 3.1f };              // in declaration order
struct Student c = { .name = "Carmen", .id = 1003 };    // by name
```

The second is a **designated initializer**: order does not matter, and anything you leave out is zero. Reorder the struct later and it still means the same thing. Prefer it.

> **Tip**
>
> Assigning one struct to another copies the whole thing — the `char` array inside it included. This is the one place C copies an array for you.

Structs Through a Pointer

p->i is (*p).i, spelled without the parentheses



"." binds tighter than "*", so *p.i would try to dereference p.i. That is why -> exists.

```
struct Point *p = &pt;

p->x        // the field x of the struct p points at
(*p).x      // the same thing, written out
```

Passing One to a Function

```
void move_by_value(struct Point p, double dx)   { p.x  += dx; }  // a copy
void move_by_pointer(struct Point *p, double dx) { p->x += dx; } // the original
```

```
sizeof(struct Point) = 24   sizeof(pointer) = 8

after move_by_value    (1.0, 2.0) tag 7
after move_by_pointer  (101.0, 2.0) tag 7
```

Passing by value copies all 24 bytes and throws the result away. Pass a pointer, and write `const struct Point *` when the function only reads.

Structs in Memory

Now the part that surprises people. Add up the fields:

```c
struct S {
    char  c;   // 1 byte
    int   i;   // 4 bytes
    short s;   // 2 bytes
};
```

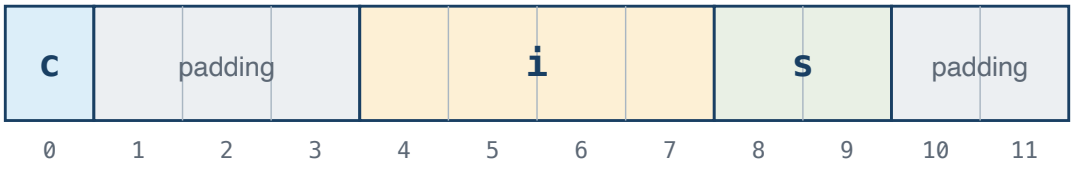
That is 7. Now ask the compiler:

```c
printf("%zu\n", sizeof(struct S));  // 12 on our x86-64 teaching systems
```

Five bytes went somewhere. The rest of this lecture is about where, and why.

Where the Five Bytes Went

Three fields that need 7 bytes take 12



The grey bytes are **padding**: nothing is stored in them.

They are there so each field starts at an offset it is allowed to start at.

Two rules produced that picture:

1. Each field starts at a multiple of its **alignment**.
2. The total rounds up to a multiple of the largest alignment in the struct, so that in an array every element stays aligned too.

Alignment Requirements

On our x86-64 teaching systems:

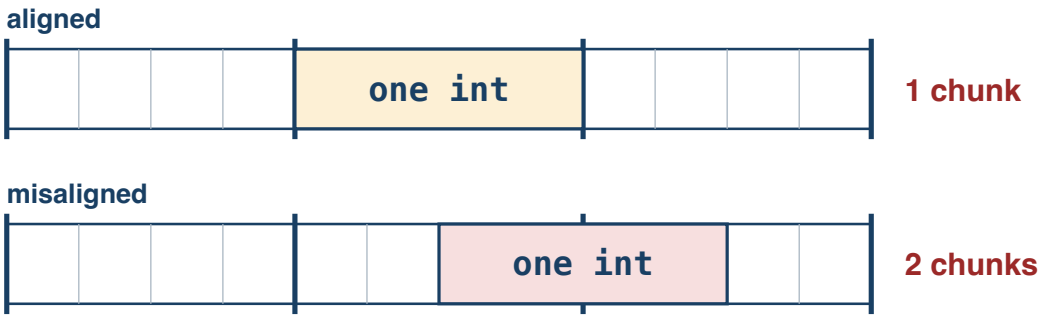
Type	Size	Must start at a multiple of
`char`	1	1 (anywhere)
`short`	2	2
`int`	4	4
`double`	8	8
pointer	8	8

Alignment is not size: `char buf[8]` has size 8 and alignment 1. Ask `_Alignof(type)` when you need the exact figure.

> **Warning**
>
> These numbers are this platform's ABI, not guarantees in the C standard. Print them on the machine you are actually targeting.

Why the Machine Cares

The machine fetches memory in fixed-size chunks



The thick lines are chunk boundaries. Our x86-64 machines will read the bottom one anyway, but it is more work — and the ABI lays structs out so it does not come up.

Memory arrives in fixed-size chunks. An aligned `int` is one chunk; a misaligned one is two, plus the work of gluing the halves. Some architectures do not offer the choice — they fault.

Walking the Rule by Hand

Take the fields in order and apply the two rules. For `struct Bad { char c; int i; short s; }`:

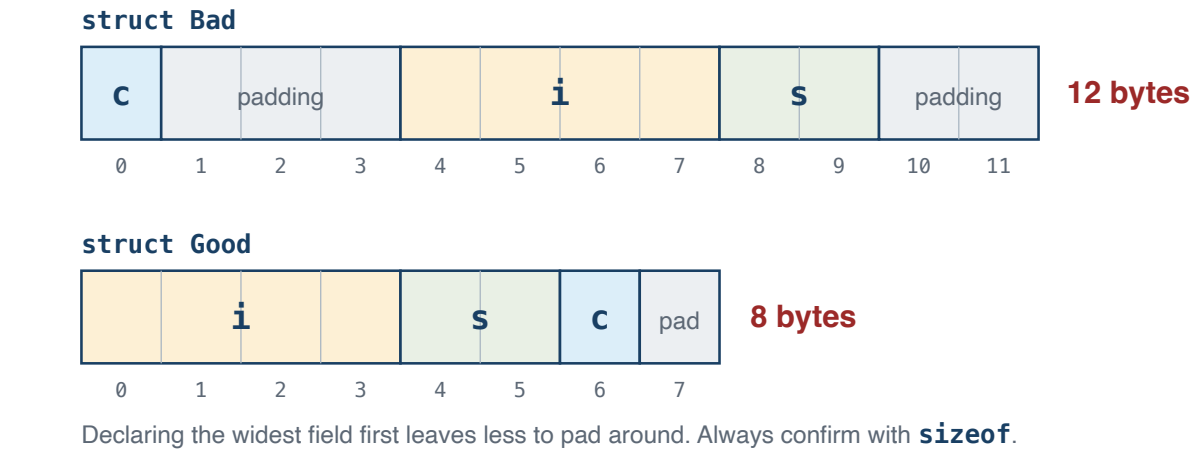
Field	Alignment	Starts at	Occupies
`c`	1	0	byte 0
`i`	4	4 — bytes 1, 2, 3 are skipped	bytes 4–7
`s`	2	8	bytes 8–9

Ten bytes so far. The largest alignment in the struct is 4, so the total rounds up to **12**.

Now do `struct Good { int i; short s; char c; }` yourself: the fields land at 0, 4 and 6, which is seven bytes, rounded up to **8**.

Reordering Saves Space

```c
struct Bad  { char c; int i; short s; };   // 12 bytes here
struct Good { int i; short s; char c; };   //  8 bytes here
```



Same three fields, same values. Declaring the widest first is the rule of thumb, and it is right most of the time.

The `offsetof` Macro

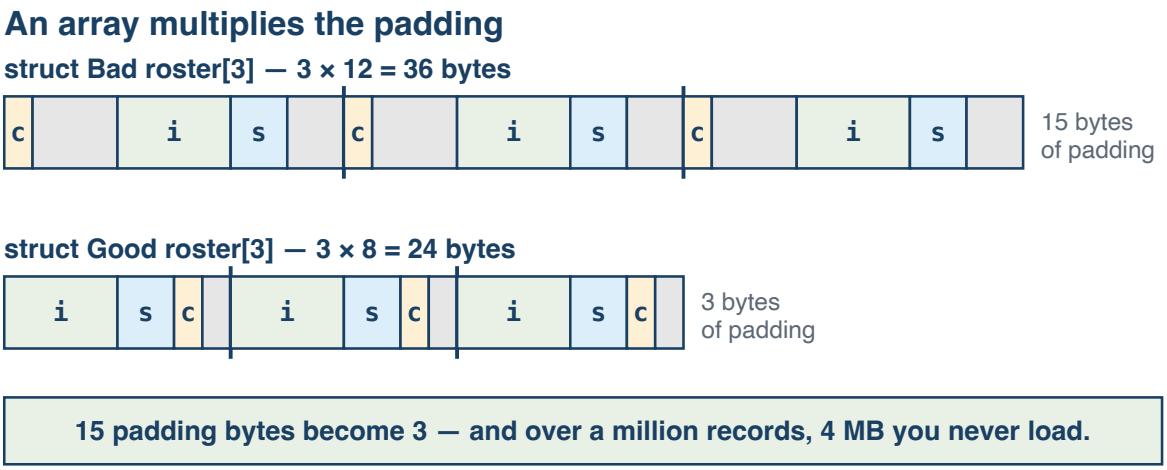
You do not have to trust your arithmetic. `offsetof` asks the compiler where a field actually landed:

```
#include <stddef.h>
offsetof(struct S, c);   // 0
```

```
struct Bad   sizeof 12  align 4        struct Good  sizeof  8  align 4
  c at 0                                 i at 0
  i at 4                                 s at 4
  s at 8                                 c at 6
```

The same three numbers you worked out by hand. Four bytes saved per record by typing the same fields in a different order — and when a struct has to match a file format, this is how you check it.

An Array Multiplies It




```
1000000 records of Bad   12000000 bytes
1000000 records of Good   8000000 bytes
```

```
difference               4000000 bytes
```

This is why the *trailing* padding exists: without it element 1 would start at offset 10, and its `int` would be misaligned.

Structs Inside Structs

A struct inside a struct is not flattened: the inner one keeps its alignment and hands it outward. And a struct cannot contain *itself* — that would need infinite space — so it holds a pointer instead, 8 bytes whatever it points at:

```
struct Inner { char c; int i; };                        // alignment 4
struct Outer { char tag; struct Inner in; char end; };
struct Node  { int value; struct Node *next; };         // alignment 8
```

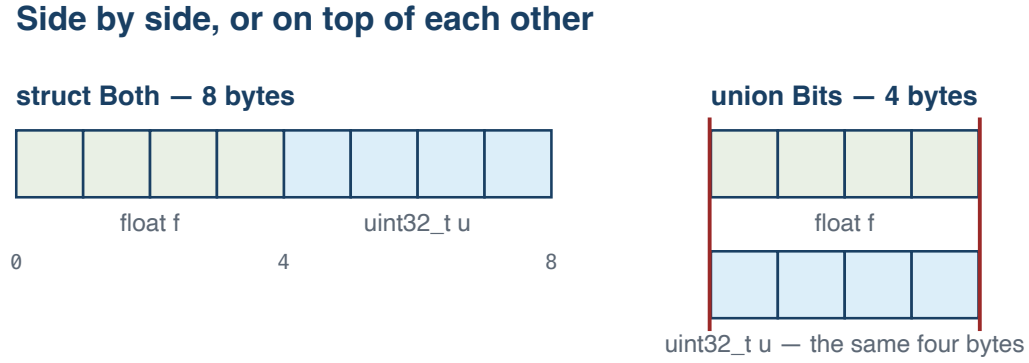
```
Inner  sizeof  8  align 4   c at 0, i at 4
Outer  sizeof 16  align 4   tag at 0, in at 4, end at 12
Node   sizeof 16  align 8   value at 0, next at 8
```

`in` starts at 4, not 1: a struct's alignment is the largest one anywhere inside it, however deep. `struct Node` is the shape of every linked list and tree you will write.

A Union Holds One Member at a Time

A struct is *all of these*. A union is *one of these*:

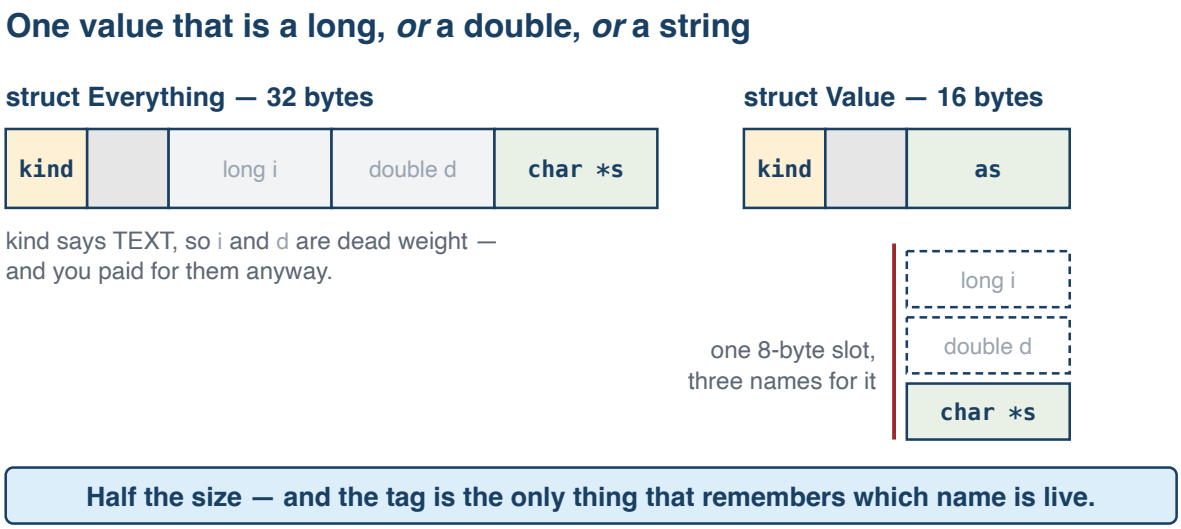
```
struct Both { float f; uint32_t u; };   // both, side by side — 8 bytes
union  Bits { float f; uint32_t u; };   // one or the other   — 4 bytes
```



Every union member starts at offset 0. Write one, read another, and you see the same bytes a second way.

Every member starts at offset 0, and the size is settled at **compile time** — the largest member, rounded up to its alignment. It never shrinks to fit: a union of a `char` and a `double` is 8 bytes while it holds the `char`. What is *not* fixed, and not recorded anywhere, is which member is live.

Why You Would Want That




```
struct Value {
    enum { V_INT, V_REAL, V_TEXT } kind;      // which member is live
    union { long i; double d; char *s; } as;
};
```

The union saves the space; the **tag** remembers what is in it. That pairing is a *tagged union* — how C says "one of these". Interpreters, JSON parsers and protocol decoders are all built on it.

Read the Member the Tag Names

```
switch (v.kind) {
case V_INT:  printf("%ld\n", v.as.i);  break;
case V_REAL: printf("%g\n",  v.as.d);  break;
case V_TEXT: printf("%s\n",  v.as.s);  break;
}
```

Read the wrong one and C says nothing at all:

```
wrote u = 42, read f -> 5.88545e-44   (nobody stopped us)
```

That is 42 as integer bits, read as a `float`. The tag is not decoration — it is the only thing standing between you and that number.

Reach for a union when the alternatives are mutually exclusive and you have many of them. `union sigval` in `<signal.h>` is one: a signal carries either an `int` or a pointer, never both.

The Other Use: The Same Bytes, Read Twice

Sometimes reading the other member is exactly what you want:

```
union Bits { float f; uint32_t u; uint8_t b[4]; };

union Bits v;
v.f = 3.0f;
```

```
v.f = 3   v.u = 0x40400000
bytes, lowest address first: 00 00 40 40
```

Well defined in C, and how Lecture 2's endianness test is usually written.

Tip

All of them at once? **struct**. One at a time? **union** plus a tag. The same bytes two ways on purpose? **union** — or `memcpy`, which also works in C++.

When the Layout Is Not Yours to Choose

So far padding has cost you memory. Here is where it costs you the answer.

Whenever a struct must match a layout **someone else defined** — a file header, a network packet, a device register — their rules and the compiler's disagree, and the compiler wins. A BMP header is the smallest real example.

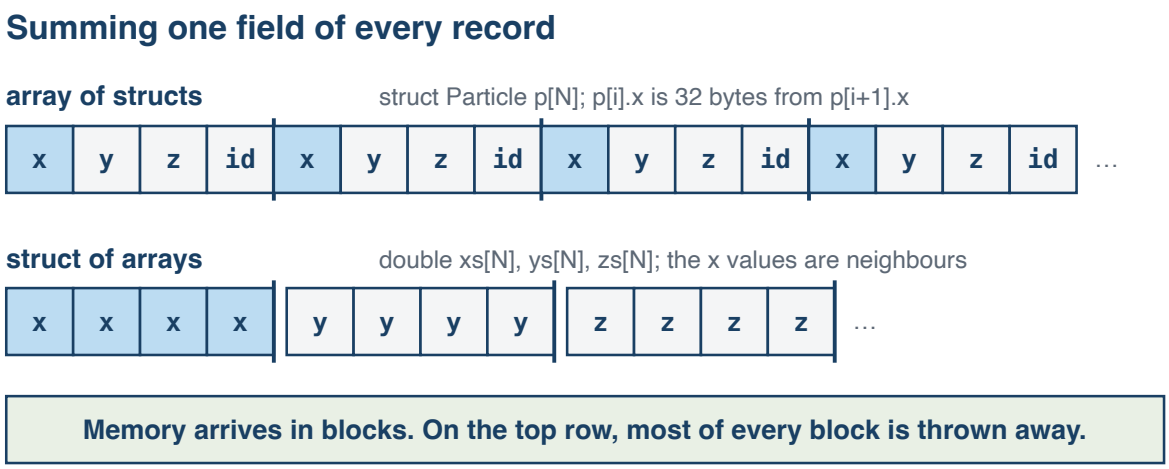
```
the file's header is  14 bytes
struct BmpNaive is    16 bytes   size at offset 4   <- the file says 2

read into BmpNaive:  size = 0    (wrong)
read byte by byte:   size = 58   offset = 54
```

`packed` fixes the size and hands you unaligned fields instead — slower, a fault on some machines, still silent about byte order. Bit fields are no better (`08-bitfields.c`). Assemble the numbers yourself:

```
uint32_t size = (uint32_t)p[0]       | (uint32_t)p[1] << 8
              | (uint32_t)p[2] << 16 | (uint32_t)p[3] << 24;
```

AI Systems Connection




```
summing x over 8000000 particles, 32 bytes each

array of structs   13.0 ms    256 MB pulled in
struct of arrays    9.3 ms     64 MB pulled in
```

Array of structs keeps a record together; **struct of arrays** keeps a field together. Every tensor library and every columnar database picks the second — they all read one field of every record. Four times the bytes, only 1.4× the time: the prefetcher hides part of a regular stride. *Lecture 15.*

Your Turn

```
struct Wasteful {
    char   a;
    double b;
```

```
    char   c;
    int    d;
    short  e;
};
```

1. What is `sizeof(struct Wasteful)`? Work it out field by field.
2. Reorder the fields to make it as small as you can. What is the new size?
3. Check both with `sizeof` and `offsetof` before you look at anyone else's answer.

Practice

1. Draw the byte layout of `struct { short a; char b; };` and predict its `sizeof`. Then compile and check.
2. Why does `struct { char c; }` have size 1, but `struct { char c; int i; }` have size 8 rather than 5?
3. Write a function that takes `struct Student *` and returns the GPA. Then write one that takes `struct Student` by value. Which one can change the caller's copy, and why?
4. `08-bitfields.c` declares an IPv4 header with bit fields. Build it. Why is `sizeof` 4 rather than 16, and why does the 16-bit `length` field come out backwards from what the wire format specifies?
5. A `struct Value` holds a tag and a union of `long`, `double` and `char *`. What is its `sizeof`, and where does the union start? Predict, then check.
6. Use a `union` to print the four bytes of `1.0f` in memory order. Which end does the exponent come out at, and why?

Reading

- CS:APP §3.9 (Heterogeneous Data Structures)